

A Tiddlywiki-Based Framework for LaTeX Publication Management and LLM-Enhanced Document Processing

Anonymous Submission

November 11, 2025

Abstract

This paper presents a novel document management framework that leverages Tiddlywiki’s transclusion architecture to decompose LaTeX publications into atomic, reusable components. The system implements a BibTeX-first organizational principle where each document element—paragraphs, equations, figures, and citations—exists as an independent tiddler with dual field representations for web rendering and LaTeX export. We specifically address the integration of Large Language Model (LLM) operations through structured quality control pipelines, automated translation systems, and OCR enhancement loops. The architecture enables fine-grained Retrieval-Augmented Generation (RAG) embeddings at the element level while maintaining full bidirectional conversion between PDF and Tiddlywiki formats. We discuss the framework’s advantages for AI-driven workflows and outline a vision for self-enforcement learning in document parsing.

1 Introduction

Traditional LaTeX-based publication workflows suffer from monolithic document structures that impede modern AI-assisted processing. Documents exist as static files, preventing granular operation on individual components and complicating integration with LLM-based tools for translation, enhancement, and retrieval. Conversely, web-based platforms lack robust mathematical typesetting and cannot easily export to standard academic formats.

Our framework resolves this tension by treating Tiddlywiki as a semantic intermediate representation. Each publication becomes a document family sharing a BibTeX key prefix, with elements stored as separate tiddlers that maintain both a KaTeX-renderable version for browser display and a pristine LaTeX field for export. This decomposition facilitates automated quality verification, multi-modal AI processing, and element-level version control while preserving the ability to reconstruct complete LaTeX projects.

2 System Architecture

2.1 BibTeX-First Organizational Principle

The framework centers on a master tiddler that functions as both a bibliographic entry and a project hub. All associated elements—paragraphs, equations, figures, and bibliographic records—bear the BibTeX key as a prefix, creating a coherent namespace. Tags categorize elements by type and provenance, while custom fields encode metadata required for rendering and export.

2.2 Dual-Field Paradigm

Each content tiddler maintains two distinct fields:

- **text**: Contains WikiText with macro-expanded mathematical expressions suitable for KaTeX rendering and templated transclusions for images and references.
- **latex**: Stores the original LaTeX code preserving all formatting commands, enabling faithful reconstruction of the source document.

This separation allows simultaneous optimization for web display and print-ready export without compromising either representation.

2.3 Templated Transclusion System

Content insertion employs a template-based transclusion mechanism of the form `{{TiddlerName||TemplateName}}`. Templates encapsulate rendering logic for different element types, ensuring consistent display across the wiki while abstracting implementation details. This approach standardizes how formulas, images, citations, and cross-references appear in the browser interface.

3 Key Features for LLM Integration

3.1 OCR Quality Control Layer

MathPix OCR processing generates coordinate metadata for each document element. For mathematical expressions, we construct tiddlers containing the LaTeX code, a KaTeX-rendered version, and a canonical URI linking to the original PDF snippet. This triple-view representation enables efficient human verification and creates training data for improving OCR accuracy. Verified elements carry a boolean flag, allowing progressive enhancement of the document corpus.

3.2 Automated Translation Pipeline

The atomic paragraph structure facilitates machine translation without affecting embedded elements. The translation service operates on the `text` field, storing the original version in an `original_text` field. This preserves provenance and enables translation rollback. The templated transclusion system ensures that mathematical expressions and figures remain intact during language conversion.

3.3 LLM-Based OCR Enhancement

Structured LLM prompts improve initial OCR quality for complex elements. TikZ diagrams, commutative diagrams, and intricate tabular structures undergo targeted extraction where the model receives the image URI and returns refined LaTeX code. This enhancement loop can be automated for large document collections, with human verification gates for critical elements.

3.4 Bibliographic Enrichment

Bibliography entries initially populated from OCR are processed through LLM queries to complete missing BibTeX fields. The system maintains a sortable bibliography table tiddler that preserves the original citation order while enriching metadata, enabling proper academic referencing in exported documents.

4 Advantages for AI-Driven Workflows

Fine-Grained Retrieval-Augmented Generation Decomposing documents into element-level tiddlers allows precise RAG embeddings for each paragraph, formula, and figure. Retrieval systems can identify relevant mathematical expressions or visual elements independently of their source documents, improving context relevance for LLM queries.

Version Control at Element Resolution Traditional LaTeX version control operates at file granularity. Our framework enables independent versioning of paragraphs and formulas, facilitating collaborative editing and precise change tracking. Each element’s modification history remains isolated, simplifying review processes.

Provenance and Confidence Tracking Every tiddler maintains provenance metadata including OCR confidence scores, LLM processing history, and verification status. This audit trail supports reproducible AI-assisted document creation and identifies elements requiring human review.

Multi-Modal AI Processing The coordinate-aware image system permits LLMs to process document regions in context. Models can reference specific page areas when generating corrections or enhancements, creating a spatially-grounded understanding of the document structure.

Embeddable Vector Indices Custom fields can store dense vector embeddings for each element, enabling similarity search across the entire publication corpus. This supports applications such as finding related equations, identifying duplicate figures, and clustering paragraphs by semantic content.

5 Future Work: Self-Enforcement Learning

5.1 Self-Improving PDF Parsing

The ultimate objective is a closed-loop system where PDF conversion quality improves automatically. Initial OCR results undergo LLM enhancement and human verification. Verified elements serve as ground truth for fine-tuning specialized vision-language models on the document corpus’s specific notation and formatting conventions. Metrics derived from verification rates and correction patterns guide active learning, prioritizing document regions where the model exhibits low confidence.

5.2 RAG Embedding Automation

Future extensions will generate embedding vectors for each tiddler automatically upon creation or modification, storing them in dedicated fields optimized for similarity search. The system will maintain vector indices for different element types (text, math, images) and support hybrid retrieval strategies combining semantic similarity with bibliographic metadata.

5.3 Collaborative Verification Interface

A specialized dashboard will aggregate elements requiring verification, prioritized by OCR confidence and citation frequency. Multiple reviewers can validate elements in parallel, with disagreements escalated to domain experts. This crowdsourced approach scales quality control for large-scale document digitization projects.

6 Conclusion

The proposed framework transforms LaTeX publications from static documents into dynamic, AI-processable knowledge graphs. By decomposing content into verified, semantically-typed tiddlers with dual representations, we enable seamless integration of LLM capabilities while preserving academic formatting standards. The architecture supports progressive enhancement through human verification and automated improvement loops, positioning it as a foundation for next-generation scholarly document management systems.